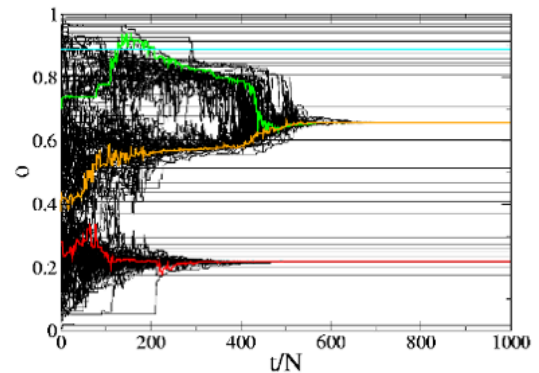


L11: Consensus Formation in Adaptive Networks

(A)

1. Choose a node randomly, node i .
2. Pick one of its neighbors, node j .
3. Generate a random number, r , between 0 and 1.
4. if ($r > w$)
 - then if ($|o(i, t) - o(j, t)| < d$)
 - then opinion convergence for i and j :
$$o(i, t+1) = o(i, t) + \mu(o(j, t) - o(i, t))$$
$$o(j, t+1) = o(j, t) - \mu(o(j, t) - o(i, t))$$
 - fi
 - else if ($|o(i, t) - o(j, t)| > d$)
 - then update the link between i and j :
 - (a) Choose a random node, node k , which is neither i nor i 's neighbor.
 - (b) Break the link between i and j then connect i with k .
 - fi
- fi
5. Start again from Step 1.

(B)



(C)

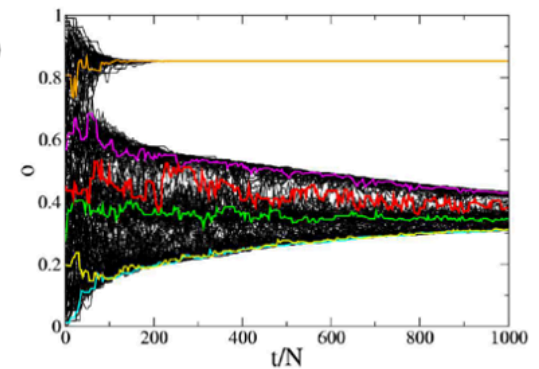


Image Source: "[Consensus formation on adaptive networks](https://arxiv.org/pdf/0707.4416.pdf)" $\Rightarrow$ (<https://arxiv.org/pdf/0707.4416.pdf>) by Kozma and Barrat.

***Correction:** The indentation of the second else if should match the first if statement.

In the following, we describe a consensus formation model for adaptive networks that was proposed by Kozma and Barrat. The model is based on the Deffuant model that we studied in Lesson-10.

N agents are endowed with a continuous opinion $o(t)$ that can vary between 0 and 1 and is initially random.

Two agents, i and j can communicate with each other if they are connected by a link.

Two neighboring agents can communicate if their opinions are close enough, i.e., if $|o(i, t) - o(j, t)| < d$, where d is the tolerance range or threshold.

In this case, the communication tends to bring the opinions even closer, according to the following "update rule":

$$o(i, t+1) = o(i, t) + \mu(o(j, t) - o(i, t))$$

and

$$o(j, t+1) = o(j, t) - \mu(o(j, t) - o(i, t))$$

where $\mu \in [0, 1/2]$ is a convergence parameter. Consider the case of $\mu = 1/2$ that corresponds to i and j adopting the same intermediate opinion after communication.

The model considers two coexisting dynamical processes: local opinion convergence for agents whose opinions are within the tolerance range, and a rewiring process for agents whose opinions differ more.

The relative frequencies of these two processes is quantified by the parameter $w \in [0, 1]$.

At each time step t , a node i and one of its neighbors j are chosen at random.

With probability w , an attempt to break the connection between i and j is made: if $|o(i, t) - o(j, t)| > d$, a new node k is chosen at random and the link (i, j) is rewired to (i, k) .

With probability $1 - w$ on the other hand, the opinions evolve according to the previous update rule. if they are within the tolerance range.

If $w > 0$, the dynamics stop when no link connects nodes with different opinions. This can correspond either to a single connected network in which all agents share the same opinion, or to several clusters representing different opinions.

For $w = 0$ on the other hand, the dynamics stops when neighboring agents either share the same opinion or differ of more than the tolerance d .

The model is described with the pseudocode shown in Figure (A).

Figures (B) and (C) show simulation results for $N=1000$ agents, a tolerance $d=0.15$, and average node degree $\bar{k} = 5$.

Figure (B) refers to the case of a static network without any rewiring, while Figure (C) refers to the adaptive model we described above (with $w=0.7$).

The evolution of the opinion of a few individuals is highlighted with color.

When the interaction network is static, local convergence processes take place and lead to a large number of opinion clusters in the final state, with few macroscopic size opinion clusters and many small size groups: agents with similar opinions may be distant on the network and not be able to communicate.

Figure (C), which corresponds to an adaptive network is strikingly in contrast with the static case: no small groups are observed.

The study of Kozma and Barrat showed the following results:

In the case of the static interaction network, two transitions are found: at large tolerance values, a global consensus is reached. For intermediate tolerance values, we see the coexistence of several extensive groups or clusters of agents sharing a common opinion with a large number of small (finite-size) clusters. At very small tolerance values finally, a fragmented state is obtained, with an extensive number of small groups.

In the case of the adaptive interaction network, i.e. when agents can break connections with neighbors with far apart opinions, the situation changes in various ways.

At large tolerance values, the polarization transition is shifted since rewiring makes it easier for a large connected cluster to be broken in various parts. The possibility of network topological change, therefore, renders global consensus more difficult to achieve.

On the other hand, for smaller tolerance values, the number of clusters is drastically reduced since agents can more easily find other agents with whom to reach an agreement. A real polarized phase is thus obtained, and the transition to a fragmented state is even suppressed: extensive clusters are obtained even at very low tolerance.

Food For Thought

Many people believe that online social networks such as Twitter lead to extreme polarization and the formation of "echo chambers". How would you explain this phenomenon using the previous model?